

Testing Pack

1. Test Environment and Accounts

- **Test Environment:** Desktop Java application running with Swing UI
- **Databases:** Auth DB and ERP DB are connected and seeded
- **Default Accounts (to be used):**
 - **Admin:** admin1
 - **Instructor:** inst1
 - **Student 1:** stu1
 - **Student 2:** stu2

2. Acceptance Tests (Must Pass)

A. Login & Roles

Test Case	User	Expected Result	Source Check
Login with correct credentials.	admin1	Dashboard matching Admin role opens	Auth DB & ERP DB
Login with wrong password.	stu1	Login is rejected; "incorrect username or password" message shown.	Auth DB hash verification

B. Student (stu1)

Test Case	Action/Data	Expected Result	Source Check
View Course Catalog.	Course: CS101	Catalog lists course details (code, title, credits, capacity, instructor)	ERP DB tables
Register (Success).	Section: MTH201	Success message; section appears in "My Registrations".	Enrollment table updated

Register (Duplicate Block).	Section: MTH201 (already registered)	Blocked with a clear message	Enrollment integrity check
Register (Section Full Block).	Section: CSE101 (capacity: 10, enrolled: 10)	Blocked with "Section full" message	Service layer check on capacity
Drop Section (Success).	Section: MTH201 (before deadline)	Success message; section is removed from registrations.	Enrollment status updated/removed
View Grades.	Registered course	Shows assessment components and final grade	Grades table data
Download Transcript.	N/A	Generates and opens a CSV file listing completed courses/grades	Util package export

C. Instructor (inst1)

Test Case	Action/Data	Expected Result	Source Check
View Assigned Sections.	N/A	Only sections assigned to inst1 are listed	Instructor access rule

Enter Scores.	Student: stu1, Section: CSE101, Component: Midterm	Score is saved successfully	Grades table updated
Compute Final Grade.	Section: CSE101	Final grade for all students is calculated and displayed using the defined weighting rule	Service compute logic

D. Admin (admin1)

Test Case	Action/Data	Expected Result	Source Check
Create New Student.	Username: stu3, Role: Student	User created in Auth DB; profile created in ERP DB (students table)	Linking both DBs via user_id
Toggle Maintenance ON.	N/A	Visible banner shows in UI; maintenance flag in settings table is true	Settings table read/write
Register while Maintenance ON.	Student: stu1	Attempt to register is blocked with a message	Service layer check on maintenance flag
Toggle Maintenance OFF.	N/A	Banner disappears; flag in settings table is false; normal student/instructor behavior returns	Service layer updates settings.

3. Edge and Negative Tests

Test Case	Action/Data	Expected Result	Source Check
Invalid Input.	Admin tries to set Section Capacity to -5 .	Input validation prevents saving; error message shown	Safe Inputs/Validation
Deadline Block.	stu1 tries to Drop a section after the stated deadline.	Blocked with a message.	Service layer logic check
Maintenance Block Scope.	Admin verifies that both student registration and instructor grade entry are blocked when Maintenance is ON.	All write operations for Student/Instructor are blocked	Service layer checks on all write paths

Small Test Dataset

Table	Field	admin1	inst1	stu1	stu2
users_auth	username, role, password	admin1, Admin, <i>password123</i>	inst1, Instructor, <i>password123</i>	stu1, Student, <i>password123</i>	stu2, Student, <i>password123</i>
students/instructors	user_id, details		User ID 2, Dept: CSE	User ID 3, Roll: 2024207	User ID 4, Roll: 2024246

courses	code, title, credits	CSE101 , Intro to Programming, 4	MTH201 , Calc I, 4	CSE404 , Adv Topics, 4	
sections	section_id, course_id, instructor_id, capacity	101, CSE101, Inst ID 2, 20	102, MTH201, Inst ID 5 (other), 15	103, CSE404, Inst ID 6 (other), 10	

Test Summary

Section	Status (Passed/Failed/N/A)	Notes/Comments (Briefly state success or known issue)
Login & Roles	Passed	All users logged in correctly; invalid credentials rejected.
Student Flows	Passed/Failed	(Section Full) passed. (Download Transcript) Failed - PDF library setup issue.
Instructor Flows	Passed	inst1 successfully entered scores and computed final grades. Access rules for non-assigned sections enforced.
Admin Flows	Passed	User creation successful. Maintenance Mode toggle and banner display works. All write actions blocked when ON.

Edge & Negative Tests	Passed/Failed	(Capacity negative input) failed to validate. Fixed by adding constraint to domain object.
Security Basics (Password/Auth)	Passed	Verified Auth DB contains bcrypt hashes only; ERP DB has no passwords.
Overall Stability/UI	Passed	Application did not crash during testing. Tables are sortable.

Conclusion: The core functionality for all three roles and the enforcement of the access rules/Maintenance Mode have passed acceptance criteria.

Login & roles

- ☒ Wrong password is rejected; shows "incorrect username or password."
- ☒ After login, the dashboard matches the role (student/instructor/admin).

Student

- ☒ See catalog with code/title/credits/capacity/instructor.
- ☒ Register in a section with free seats → success and appears in "My Registrations" and timetable.
- ☒ Try to register the *same* section again → blocked with a clear message.
- ☒ Try to register in a *full* section → blocked with "Section full."
- ☒ Drop before deadline → success; appears removed.
- ☒ View grades for registered courses and sections.
- ☒ Download transcript (CSV or PDF) listing completed courses/grades.

Instructor

- ☒ See only their own sections.
- ☒ Enter scores (quiz/midterm/end-sem) → saved.
- ☒ Compute final grade using the rule in the report → shows final values.

- ☐ Try to edit a section they do not teach → blocked with "Not your section."

Admin

- ☐ Create a new student user (username+role in Auth DB; student profile in ERP DB).
- ☐ Create a course and a section; assign instructor.
- ☐ Toggle Maintenance ON → banner shows; student/instructor can *view* but *cannot change* (register/drop, grades blocked with message).
- ☐ Toggle Maintenance OFF → normal behavior returns.

Password & Auth separation

- ☐ Password *hash* exists in Auth DB (not the real password).
- ☐ ERP DB does not contain passwords.
- ☐ Login uses Auth DB, then loads student/instructor profile from ERP DB via the shared `user_id`.

Exports (pick at least one)

- ☐ Transcript export (Student) *or* grade CSV export (Instructor) works and opens.

C. Edge & negative tests

- ☐ Capacity cannot be negative or nonsensical.
- ☐ Register/drop after the stated deadline → blocked with message.
- ☐ Student cannot view or change another student's data.
- ☐ Instructor cannot grade students from another instructor's section.
- ☐ With Maintenance ON, student/instructor changes are blocked everywhere (not only on one screen).

D. Data integrity checks

- ☐ Duplicate enrollments (same student+same section) are prevented.
- ☐ Removing a section: either blocked when students are enrolled or clearly explained/documented.

E. Security basics

- ☐ Real passwords are never stored; only hashes.
- ☐ A Change Password dialog exists (bonus).
- ☐ After 5 wrong login attempts, show a warning or temporary lock (bonus).
- ☐ Access rules checker is called before any change (we will try to bypass via UI flow).

F. UI/UX checks

- ☒ Buttons and labels are clear; errors are friendly.
- ☒ Tables for lists; sorting/filtering is a plus.
- ☒ Long actions show "please wait" (even a simple dialog).

G. Performance sanity

- ☒ Catalog list of ~100 courses opens quickly (a few seconds or less).
- ☒ App starts without long hangs or crashes.

H. Maintenance & (optional) backup/restore

- ☒ Maintenance toggle flips the settings flag and the UI shows a banner immediately.
- ☒ All student/instructor *writes* are blocked while ON.
- ☒ (Bonus) Backup/restore: trigger backup, change a row, restore, and see it revert.